

Question 1 - Intro to concurrency

- Motivation
- Types of Concurrency
- Mutual Exclusion
- Critical Sections
- Data Races vs Race Conditions

Question 2 - Synchronization

- Locks
- Monitors
- Fairness
- Semaphores ($c=1$)

Question 3 - Visibility

- Visibility
- Reordering
- Effect of Volatile / Locks
- $V \neq ME$

Question 4 - Java memory model

- Validity / Well-Formed
- Program Order
- Happens-Before Order (& Rules)
- Synchronization Order
- Conflicting Actions
- “Correctly Synchronized Program”

Question 5 - Thread-safe classes

- Definition
- Class State
- Escaping
- Safe Publication
- Immutability / Mutual Exclusion
- Instance Confinement

Question 6 - Testing

- Motivation
- Safety vs. Liveness
- Functional Correctness Tests
 - Parametrized
 - Repeated
 - Barriers
- Formal Verification

Question 7 - Performance measurements

- Motivation
- Benchmarking
 - Fluctuations
 - Mark7
- Thread Creation

Question 8 - Performance and Scalability

- Motivation
- Thread Pools
 - Splitting Tasks
 - Thread Usage
- Lock Striping

Question 9 - Lock-free Data Structures

- Motivation over locks
- Memory Overhead
- Compare-And-Swap / -Set
 - Atomic
 - Thread-Safe

Question 10 - Linearizability

- Sequential Specification
- Sequential Consistency
- Linearizability
- Linearization Points
- Concurrent Objects

Question 11 - Streams

- Functional Pipeline
- Lazy, Parallelizable & Memory
- Sources
- Intermediate Operations
- Terminal Operations

Question 12 - Message Passing

- Motivation
- Actor Model
 - State, Mailbox & Actions
- Actor model => Erlang
- Dynamic Topologies
 - Load Balancing
 - Fault Tolerance